Special Case

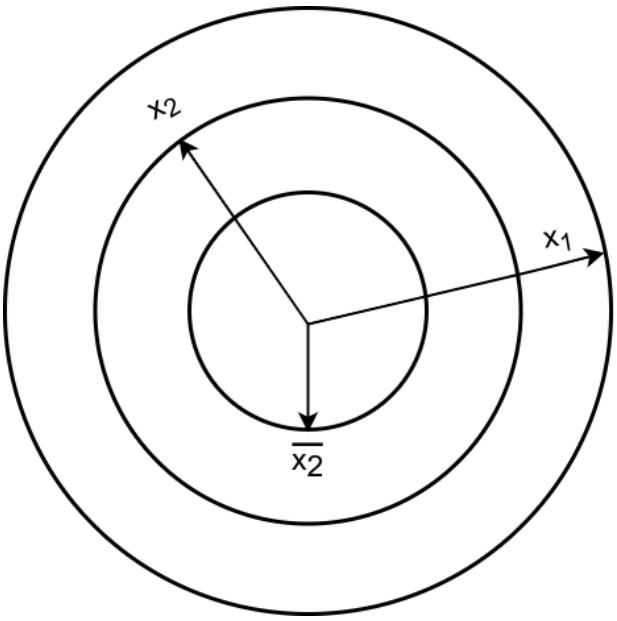


Figure 1: Distribution of quarks in three regions

The functional relationships are:

$$f_1(x) = \beta_{1\bar{2}}\overline{f_2(x)} \quad \text{(in the inner region)}$$
$$f_2(x) = \beta_{2\bar{2}}\overline{f_2(x)} \quad \text{(in the inner region)}$$
$$f_1(x) = \gamma_{12} f_2(x) \quad \text{(in the middle region)}$$

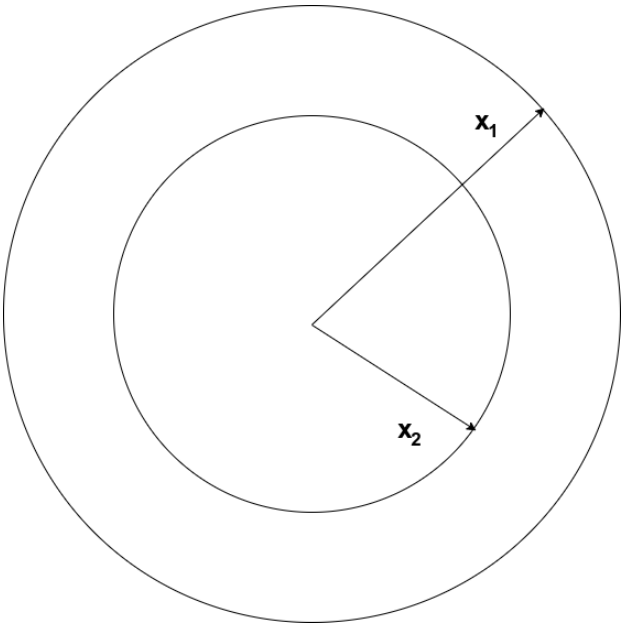


Figure 2: Distribution of quarks in two regions

The functional relationships are:

$$f_1(x) = k f_2(x) \quad \text{(in the inner region)}$$
$$\overline{f_2(x)} = \overline{k} f_2(x) \quad \text{(in the inner region)}$$

1 Kinetic Energy

$$\frac{T_1}{C}+\frac{T_2}{C}=\alpha_1 g_1\int_0^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\overline{\alpha_2 g_2}\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$=\alpha_1 g_1\int_0^{\overline{x_2}}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{\overline{x_2}}^{x_2}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_0^{\overline{x_2}}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_{\overline{x_2}}^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$
$$+\overline{\alpha_2 g_2}\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

Let,

$\overline{N_2}=N_2,\quad \overline{Q_2}=Q_2,\quad \overline{x_2}=x_2,\quad \beta_{1\overline{2}}=k,\quad \gamma_{12}=k$

But,

$[\overline{f_2(x)}=f_2(x),\quad \beta_{2\overline{2}}=1]\quad$ for $\quad \overline{x_2}\le x\le x_2$

$[\overline{f_2(x)}=\overline{k}f_2(x),\quad \beta_{2\overline{2}}=\overline{k}]\quad$ for $\quad 0\le x\le \overline{x_2}$

$$=\alpha_1 g_1\int_0^{x_2}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_2}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_{x_2}^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$
$$+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\left[\text{The integration }\int_{x_2}^{x_2}\text{ doesnot have any significance. So, we neglect it.}\right]$$

$$=\alpha_1 g_1\int_0^{x_2}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_2 g_2\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$=\underbrace{\alpha_1 g_1(k)^{\frac{5}{2}}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}}_{(\gamma_{12}=k)}+\alpha_2 g_2\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$=\left[\alpha_1 g_1(k)^{\frac{5}{2}}+\alpha_2 g_2+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\right]\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\alpha_1 g_1\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$=\left[\alpha_1 g_1(k)^{\frac{5}{2}}+\alpha_2 g_2+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\right]\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5N_2}{21\eta}\,\frac{df_2(x)}{dx}\Big|_{x_2}\right]+\alpha_1 g_1\left[\frac{4}{7}\sqrt{x}\,(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}(k)^{\frac{5}{2}}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}\right.$$
$$-\frac{5}{7Q_1}\,\frac{df_1(x)}{dx}\Big|_{x_1}\left\{\frac{N_1Q_1}{3\eta}+k\frac{N_2Q_2}{3\eta}-(k)^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+\frac{N_2Q_2}{Q_1}\left((k)^{\frac{3}{2}}\cdot 1-(k)^{\frac{3}{2}}\right)-k\cdot(1)^{\frac{3}{2}}\cdot\frac{N_2Q_2}{3\eta}+k\cdot 1\cdot\frac{N_2Q_2}{3\eta}\right\}$$
$$\left.+\frac{5}{7Q_1}(k)^2\,\frac{df_2(x)}{dx}\Big|_{x_2}\left\{\frac{N_2Q_2}{3\eta}+1\cdot\frac{N_2Q_2}{3\eta}-(1)^{\frac{3}{2}}\frac{N_2Q_2}{3\eta}\right\}\right]$$

$$=\frac{4}{7}\alpha_1 g_1(k)^{\frac{5}{2}}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}+\frac{4}{7}\alpha_2 g_2\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}+\frac{4}{7}\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5N_2}{21\eta}\alpha_1 g_1(k)^{\frac{5}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$-\frac{5N_2}{21\eta}\alpha_2 g_2\,\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2}{21\eta}\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{4}{7}\alpha_1 g_1\sqrt{x}\,(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}\alpha_1 g_1(k)^{\frac{5}{2}}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$
$$-\frac{5N_1}{21\eta}\alpha_1 g_1\,\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2Q_2}{21\eta Q_1}\alpha_1 g_1 k\,\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5N_2}{21\eta}(k)^{\frac{3}{2}}\alpha_1 g_1\,\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5N_2Q_2}{21\eta Q_1}(k)^2\alpha_1 g_1\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\boxed{\begin{aligned}&=\frac{4}{7}\alpha_1 g_1\sqrt{x}\,(f_1(x))^{\frac{5}{2}}\Big|_{x_1}+\frac{4}{7}\left\{\alpha_1 g_1(k)^{\frac{5}{2}}+\alpha_2 g_2+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}-\alpha_1 g_1(k)^{\frac{5}{2}}\right\}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5N_2}{21\eta}\left[\alpha_1 g_1(k)^{\frac{5}{2}}\right.\\&\left.+\alpha_2 g_2+\overline{\alpha_2 g_2}(\overline{k})^{\frac{5}{2}}-k^2\frac{Q_2}{Q_1}\alpha_1 g_1\,\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5}{21\eta}\alpha_1 g_1\left[N_1+\frac{N_2Q_2}{Q_1}k-N_2(k)^{\frac{3}{2}}\right]\,\frac{df_1(x)}{dx}\Big|_{x_1}\right.\end{aligned}}$$

2 $K_{\overline{2}2}$

$$K_{\overline{2}2} = \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

Let,

$\overline{N_2} = N_2, \quad \overline{Q_2} = Q_2, \quad \overline{x_2} = x_2, \quad \beta_{1\overline{2}} = k, \quad \gamma_{12} = k$

But,

$[\overline{f_2(x)} = f_2(x), \quad \beta_{2\overline{2}} = 1] \quad \text{for} \quad \overline{x_2} \leq x \leq x_2$

$[\overline{f_2(x)} = \overline{k} f_2(x), \quad \beta_{2\overline{2}} = \overline{k}] \quad \text{for} \quad 0 \leq x \leq \overline{x_2}$

$$= (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \cdot \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] + (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \left[\frac{1}{Q_2} \frac{df_2(x)}{dx} \bigg|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx} \right]$$

$$= \frac{(\overline{k})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} - \frac{(\overline{k})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta} (\overline{k})^{\frac{3}{2}} \left[\frac{1}{Q_2} \frac{df_2(x)}{dx} \bigg|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx} \right]$$

$$= \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} - \frac{(\overline{k})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2} - \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx}$$

$$= - \frac{(\overline{k})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2}$$

$$= - \frac{(\overline{k})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7} \sqrt{x} (f_2(x))^{\frac{5}{2}} \bigg|_{x_2} - \frac{5N_2}{21\eta} \frac{df_2(x)}{dx} \bigg|_{x_2} \right] + \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2}$$

$$= - \frac{4}{7Q_2} (\overline{k})^{\frac{3}{2}} \sqrt{x} (f_2(x))^{\frac{5}{2}} \bigg|_{x_2} + \frac{5N_2}{21\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2} + \frac{N_2}{3\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2}$$

$$\boxed{= - \frac{4}{7Q_2} (\overline{k})^{\frac{3}{2}} \sqrt{x} (f_2(x))^{\frac{5}{2}} \bigg|_{x_2} + \frac{4N_2}{7\eta Q_2} (\overline{k})^{\frac{3}{2}} \frac{df_2(x)}{dx} \bigg|_{x_2}}$$

3 $K_{\overline{2}1}$

$$K_{\overline{2}1} = \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

Let,

$\overline{N_2} = N_2, \quad \overline{Q_2} = Q_2, \quad \overline{x_2} = x_2, \quad \beta_{1\overline{2}} = k, \quad \gamma_{12} = k$

But,

$[\overline{f_2(x)} = f_2(x), \quad \beta_{2\overline{2}} = 1] \quad \text{for} \quad \overline{x_2} \leq x \leq x_2$

$[\overline{f_2(x)} = \overline{k} f_2(x), \quad \beta_{2\overline{2}} = \overline{k}] \quad \text{for} \quad 0 \leq x \leq \overline{x_2}$

$$= (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \cdot \frac{k^{\frac{3}{2}}}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] + (\overline{k})^{\frac{3}{2}} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \left[\frac{(k)^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} \bigg|_{x_2} - \frac{(k)^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} \right.$$

$$\left. + \frac{(k)^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} - \frac{(k)^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} \bigg|_{x_2} + \frac{1}{Q_1} \frac{df_1(x)}{dx} \bigg|_{x_1} - \frac{k}{Q_1} \frac{df_2(x)}{dx} \bigg|_{x_2} \right]$$

$$= \frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} - \frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}$$
$$- \frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} + \frac{(\overline{k})^{\frac{3}{2}}}{Q_1}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{k(\overline{k})^{\frac{3}{2}}}{Q_1}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}$$

$$= -\frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{(\overline{k})^{\frac{3}{2}}}{Q_1}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1}$$
$$- \frac{k(\overline{k})^{\frac{3}{2}}}{Q_1}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}$$

$$= -\frac{(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}}{Q_2}\left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5N_2}{21\eta}\frac{df_2(x)}{dx}\Big|_{x_2}\right] + \frac{N_2}{3\eta Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}(\overline{k})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}k(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{4}{7Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta Q_1}k(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \frac{N_2}{3\eta Q_1}(\overline{k})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= -\frac{4}{7Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4N_2}{7\eta Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta Q_1}k(\overline{k})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}(\overline{k})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\boxed{= -\frac{4}{7Q_2}(k)^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{4N_2}{7\eta Q_2}k^{\frac{3}{2}}(\overline{k})^{\frac{3}{2}} - \frac{N_2}{3\eta Q_1}k(\overline{k})^{\frac{3}{2}}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}(\overline{k})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}}$$

4 K_{22}

$$K_{22} = \frac{4}{7}(\beta_{2\overline{2}})^2\left\{\frac{(\beta_{2\overline{2}})^{\frac{1}{2}}}{Q_2} - \frac{\beta_{2\overline{2}}}{\overline{Q_2}}\right\}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^2\left\{\beta_{2\overline{2}}\frac{12}{7\overline{Q_2}} + \frac{2}{7}\frac{\overline{Q_2}}{{Q_2}^2} - \frac{2(\beta_{2\overline{2}})^{\frac{1}{2}}}{Q_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \left[\frac{12N_2}{21\eta Q_2} + \beta_{2\overline{2}}\frac{\overline{N_2}}{3\eta Q_2}\left\{(\beta_{12})^{\frac{1}{2}} - \frac{2\overline{Q_2}}{7Q_2} - \frac{5}{7}(\beta_{2\overline{2}})^{\frac{1}{2}}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

Here,

$\overline{N_2} = N_2, \quad \overline{Q_2} = Q_2, \quad \overline{x_2} = x_2, \quad \overline{f_2(x)} = f_2(x), \quad \beta_{2\overline{2}} = 1$

$$= \frac{4}{7}(1)^2\left\{\frac{(1)^{\frac{1}{2}}}{Q_2} - \frac{1}{Q_2}\right\}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{N_2}{3\eta}(1)^2\left\{\frac{1\cdot 12}{7Q_2} + \frac{2}{7Q_2} - \frac{2}{Q_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \left[\frac{12N_2}{21\eta Q_2} + \frac{N_2}{3\eta Q_2}\left\{1 - \frac{2}{7} - \frac{5}{7}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{N_2}{3\eta}\left\{\frac{2}{Q_2} - \frac{2}{Q_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{12N_2}{21\eta Q_2} + \frac{N_2}{3\eta Q_2}\left\{\frac{7-7}{7}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{12N_2}{21\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\boxed{= -\frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4N_2}{7\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}}$$

5 K_{21}

$$K_{21} = \frac{4}{7}(\beta_{2\overline{2}})^{\frac{3}{2}}\left\{\frac{(\gamma_{12})^{\frac{3}{2}}\beta_{2\overline{2}}}{Q_2} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\right\}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}\beta_{2\overline{2}}\left\{\frac{5}{7\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{1}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\right.$$
$$\left. - \frac{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2} - \frac{5\overline{Q_2}}{7Q_2^2}(\gamma_{12})^{\frac{3}{2}}\beta_{2\overline{2}} + \frac{\overline{Q_2}}{Q_2^2}(\gamma_{12})^{\frac{3}{2}}\beta_{2\overline{2}} + \frac{1}{\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{1}{2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{Q_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{1}{3\eta Q_2}\left[N_2(\gamma_{12})^{\frac{3}{2}} - \frac{N_2Q_2}{Q_1}\gamma_{12} + \overline{N_2}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{Q_2}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}}) + \frac{5}{7}(\gamma_{12})^{\frac{3}{2}}\left\{N_2 + \frac{\overline{N_2Q_2}}{Q_2}\beta_{2\overline{2}}\right.\right.$$
$$\left.\left. - \overline{N_2}(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

Here,

$$\overline{N_2} = N_2, \quad \overline{Q_2} = Q_2, \quad \overline{x_2} = x_2, \quad \overline{f_2(x)} = f_2(x), \quad \beta_{1\overline{2}} = k \quad \beta_{2\overline{2}} = 1, \quad \gamma_{12} = k$$

$$= \frac{4}{7}(1)^{\frac{3}{2}}\left\{\frac{k^{\frac{3}{2}}\cdot 1}{Q_2} - \frac{k^{\frac{3}{2}}}{Q_2}\right\}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}k^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{N_2}{3\eta}\left\{\frac{5}{7Q_2}(1)^{\frac{1}{2}}(k)^{\frac{3}{2}} - \frac{(k)^{\frac{3}{2}}(1)^{\frac{3}{2}}}{Q_2}\right.$$
$$\left. - \frac{5}{7Q_2}(k)^{\frac{3}{2}}\cdot 1 + \frac{k^{\frac{3}{2}}\cdot 1}{Q_2} + \frac{1}{Q_2}(k)^{\frac{3}{2}}\cdot(1)^{\frac{1}{2}} - \frac{k^{\frac{3}{2}}}{Q_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{3\eta Q_2}\left[N_2(k)^{\frac{3}{2}} - \frac{N_2Q_2}{Q_1}k + N_2(k)^{\frac{3}{2}} - N_2(k)^{\frac{3}{2}}\cdot 1\right.$$
$$\left. + \frac{5}{7}(k)^{\frac{3}{2}}\left\{N_2 + N_2\cdot 1 - N_2\cdot(1)^{\frac{3}{2}}\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= \frac{4}{7}\left(\frac{k^{\frac{3}{2}}}{Q_2} - \frac{k^{\frac{3}{2}}}{Q_2}\right)\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(k)^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{N_2}{3\eta}\left[\frac{5k^{\frac{3}{2}}}{7Q_2} - \frac{k^{\frac{3}{2}}}{Q_2} - \frac{5k^{\frac{3}{2}}}{7Q_2} + \frac{k^{\frac{3}{2}}}{Q_2} - \frac{k^{\frac{3}{2}}}{Q_2} + \frac{k^{\frac{3}{2}}}{Q_2}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \frac{1}{3\eta Q_2}\left[N_2k^{\frac{3}{2}} - \frac{N_2Q_2}{Q_1}k + N_2k^{\frac{3}{2}} - N_2k^{\frac{3}{2}} + \frac{5N_2}{7}k^{\frac{3}{2}} + \frac{5N_2}{7}k^{\frac{3}{2}} - \frac{5N_2}{7}k^{\frac{3}{2}}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= -\frac{4}{7Q_2}(k)^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{N_2}{3\eta Q_2}k^{\frac{3}{2}} - \frac{N_2}{3\eta Q_1}k + \frac{5N_2}{21\eta Q_2}k^{\frac{3}{2}}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\boxed{= -\frac{4}{7}(k)^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{4N_2}{7\eta Q_2}k^{\frac{3}{2}} - \frac{N_2}{3\eta Q_1}k\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}}$$

6 K_{11}

$$
\begin{aligned}
K_{11} = \frac{4}{7}\left[\frac{(\beta_{2\overline{2}})^{\frac{5}{2}}(\gamma_{12})^3}{Q_2} - \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\right]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7}\left[\frac{(\gamma_{12})^{\frac{5}{2}}}{Q_1} - \frac{(\gamma_{12})^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} \\
+ \left[\frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\beta_{2\overline{2}}}{Q_2} + \frac{\overline{Q_2}}{{Q_2}^2}(\gamma_{12})^3(\beta_{2\overline{2}})^2 - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\beta_{2\overline{2}}}{Q_2}\right\} + \frac{5\overline{N_2}}{21\eta}\left\{\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\right. \right. \\
\left. \left. - \frac{\overline{Q_2}}{{Q_2}^2}(\beta_{2\overline{2}})^2(\gamma_{12})^3\right\}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \left[\frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3}{Q_2} + \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}}}{Q_1}\right. \right. \\
\left. + \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2} - \frac{\overline{Q_2}}{{Q_2}^2}(\gamma_{12})^3\beta_{2\overline{2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1} + \frac{(\gamma_{12})^{\frac{5}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_1} + \frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{12})^2\beta_{2\overline{2}} - \frac{Q_2}{{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\right\} \\
+ \frac{N_2}{3\eta}\left\{\frac{(\gamma_{12})^3}{Q_2} - \frac{2(\gamma_{12})^{\frac{5}{2}}}{Q_1} + \frac{Q_2}{{Q_1}^2}(\gamma_{12})^2\right\} + \frac{5}{21\eta}\left\{\frac{N_2}{Q_2}(\gamma_{12})^3 + \frac{\overline{N_2Q_2}}{{Q_2}^2}(\gamma_{12})^3\beta_{2\overline{2}} - \frac{\overline{N_2}}{Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\right. \\
\left. - \frac{N_2Q_2}{{Q_1}^2}(\gamma_{12})^2 - \frac{\overline{N_2Q_2}}{{Q_1}^2}(\gamma_{12})^2\beta_{2\overline{2}} + \frac{\overline{N_2}Q_2}{{Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_1}{3\eta Q_1} - \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{12} + \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\right. \\
+ \frac{\overline{N_2}}{3\eta Q_1}\left\{(\beta_{1\overline{2}})^{\frac{3}{2}} - (\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{Q_2}}{Q_1}\beta_{2\overline{2}}\gamma_{12} + \frac{Q_2}{Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{Q_1} + \frac{N_2Q_2}{{Q_1}^2}\gamma_{12} - \frac{N_2}{Q_1}(\gamma_{12})^{\frac{3}{2}}\right. \\
\left.\left. + \frac{\overline{N_2}}{Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2}}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2}Q_2}{{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12} + \frac{\overline{N_2Q_2}}{{Q_1}^2}\gamma_{12}\beta_{2\overline{2}}\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

Here,

$$
\overline{N_2} = N_2, \quad \overline{Q_2} = Q_2, \quad \overline{x_2} = x_2, \quad \overline{f_2(x)} = f_2(x), \quad \beta_{1\overline{2}} = k \quad \beta_{2\overline{2}} = 1, \quad \gamma_{12} = k
$$

$$
\begin{aligned}
= \frac{4}{7}\left[\frac{(1)^{\frac{5}{2}}k^3}{Q_2} - \frac{k^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\left[\frac{k^{\frac{5}{2}}}{Q_1} - \frac{k^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \left[\frac{N_2}{3\eta}\left\{\frac{k^3}{Q_2} - \frac{k^{\frac{3}{2}}\cdot k^{\frac{3}{2}}\cdot 1}{Q_2}\right. \right. \\
\left. + \frac{k^3\cdot 1^2}{Q_2} - \frac{k^{\frac{3}{2}}\cdot k^{\frac{3}{2}}\cdot 1}{Q_2}\right\} + \frac{5N_2}{21\eta}\left\{\frac{k^3}{Q_2} - \frac{(1)^2\cdot k^3}{Q_2}\right\}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_2}{3\eta}\left\{\frac{k^{\frac{3}{2}}\cdot k^{\frac{3}{2}}}{Q_2} - \frac{(k)^{\frac{3}{2}}\cdot k}{Q_1} - \frac{(1)^{\frac{3}{2}}\cdot k^3}{Q_2} + \frac{(1)^{\frac{3}{2}}\cdot k^{\frac{5}{2}}}{Q_1}\right. \right. \\
\left. + \frac{(k)^{\frac{3}{2}}\cdot k^{\frac{3}{2}}}{Q_2} - \frac{k^3\cdot 1}{Q_2} - \frac{k^{\frac{3}{2}}\cdot k}{Q_1} + \frac{k^{\frac{5}{2}}\cdot(1)^{\frac{3}{2}}}{Q_1} + \frac{Q_2\cdot(k)^2\cdot 1}{Q_1^2} - \frac{Q_2}{Q_1^2}(1)^{\frac{3}{2}}\cdot k^2\right\} + \frac{N_2}{3\eta}\left\{\frac{k^3}{Q_2} - \frac{2k^{\frac{5}{2}}}{Q_1} + \frac{Q_2}{Q_1^2}k^2\right\} \\
+ \frac{5}{21\eta}\left\{\frac{N_2}{Q_2}k^3 + \frac{N_2}{Q_2}k^3\cdot 1 - \frac{N_2}{Q_2}(1)^{\frac{3}{2}}\cdot k^3 - \frac{N_2Q_2}{Q_1^2}k^2 - \frac{N_2Q_2}{Q_1^2}k^2\cdot 1 + \frac{N_2Q_2}{Q_1^2}k^2\cdot(1)^{\frac{3}{2}}\right\}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} \\
+ \left[\frac{N_1}{3\eta Q_1} - \frac{N_2Q_2}{3\eta Q_1^2}k + \frac{N_2}{3\eta Q_1}k^{\frac{3}{2}} + \frac{N_2}{3\eta Q_1}\left\{k^{\frac{3}{2}} - (k)^{\frac{3}{2}}\cdot(1)^{\frac{3}{2}} - \frac{Q_2}{Q_1}\cdot k\cdot 1 + \frac{Q_2}{Q_1}(1)^{\frac{3}{2}}\cdot k\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{Q_1} + \frac{N_2Q_2}{Q_1^2}k\right. \right. \\
\left.\left. - \frac{N_2}{Q_1}(k)^{\frac{3}{2}} + \frac{N_2}{Q_1}(k)^{\frac{3}{2}}\cdot(1)^{\frac{3}{2}} - \frac{N_2}{Q_1}(k)^{\frac{3}{2}} - \frac{N_2Q_2}{Q_1^2}(1)^{\frac{3}{2}}\cdot k + \frac{N_2Q_2}{Q_1^2}\cdot k\cdot 1\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
= \frac{4}{7}\left[\frac{k^3}{Q_2} - \frac{k^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\left[\frac{k^{\frac{5}{2}}}{Q_1} - \frac{k^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \left[\frac{N_2}{3\eta}\left\{\frac{k^3}{Q_2} - \frac{k^{\frac{3}{2}}\cdot k^{\frac{3}{2}}}{Q_2} + \frac{k^3}{Q_2}\right. \right. \\
\left. - \frac{k^{\frac{3}{2}}\cdot k^{\frac{3}{2}}}{Q_2}\right\} + \frac{5N_2}{21\eta}\left\{\frac{k^3}{Q_2} - \frac{k^3}{Q_2}\right\}\Big]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_2}{3\eta}\left\{\frac{k^3}{Q_2} - \frac{k^{\frac{5}{2}}}{Q_1} - \frac{k^3}{Q_2} + \frac{k^{\frac{5}{2}}}{Q_1} + \frac{k^3}{Q_2} - \frac{k^3}{Q_2} - \frac{k^{\frac{5}{2}}}{Q_1} + \frac{k^{\frac{5}{2}}}{Q_1} + \frac{k^2Q_2}{Q_1^2} - \frac{k^2Q_2}{Q_1^2}\right\} \right. \\
\left. + \frac{N_2}{3\eta}\left\{\frac{k^3}{Q_2} - \frac{2k^{\frac{5}{2}}}{Q_1} + \frac{k^2Q_2}{Q_1^2}\right\} + \frac{5}{21\eta}\left\{\frac{N_2k^3}{Q_2} + \frac{N_2k^3}{Q_2} - \frac{N_2k^3}{Q_2} - \frac{N_2Q_2}{Q_1^2}k^2 - \frac{N_2Q_2}{Q_1^2}k^2 + \frac{N_2Q_2}{Q_1^2}k^2\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} \\
+ \left[\frac{N_1}{3\eta Q_1} - \frac{N_2Q_2}{3\eta Q_1^2}k + \frac{N_2}{3\eta Q_1}k^{\frac{3}{2}} + \frac{N_2}{3\eta Q_1}\left\{k^{\frac{3}{2}} - k^{\frac{3}{2}} - \frac{Q_2}{Q_1}k + \frac{Q_2}{Q_1}k\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{Q_1} + \frac{N_2Q_2}{Q_1^2}k - \frac{N_2}{Q_1}k^{\frac{3}{2}} + \frac{N_2}{Q_1}k^{\frac{3}{2}}\right. \\
\left.\left. - \frac{N_2}{Q_1}k^{\frac{3}{2}} - \frac{N_2Q_2}{Q_1^2}k + \frac{N_2Q_2}{Q_1^2}k\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

$$= \frac{4}{7}\left(\frac{k^{\frac{5}{2}}}{Q_1} - \frac{k^3}{Q_2}\right)\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \left[\frac{N_2}{3\eta}\left(\frac{k^3}{Q_2} - \frac{2k^{\frac{5}{2}}}{Q_1} + \frac{k^2Q_2}{Q_1^2}\right) + \frac{5}{21\eta}\left(\frac{N_2k^3}{Q_2}\right.\right.$$
$$\left.\left. - \frac{N_2Q_2}{Q_1^2}k^2\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_1}{3\eta Q_1} - \frac{N_2Q_2}{3\eta Q_1^2}k + \frac{N_2}{3\eta Q_1}k^{\frac{3}{2}} + \frac{5N_1}{21\eta Q_1} + \frac{5N_2Q_2}{21\eta Q_1^2}k - \frac{5N_2}{21\eta Q_1}k^{\frac{3}{2}}\right]\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= -\frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}k^{\frac{3}{2}}\left(\frac{k^{\frac{3}{2}}}{Q_2} - \frac{k}{Q_1}\right)\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{N_2k^3}{3\eta Q_2} - \frac{2N_2k^{\frac{5}{2}}}{3\eta Q_1} + \frac{N_2Q_2k^2}{3\eta Q_1^2} + \frac{5N_2k^3}{21\eta Q_2}\right.$$
$$\left. - \frac{5N_2Q_2}{21\eta Q_1^2}k^2\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{4N_1}{7\eta Q_1} - \frac{2N_2Q_2k}{21\eta Q_1^2}k + \frac{2N_2k^{\frac{3}{2}}}{21\eta Q_1}\right]\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= -\frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}k^{\frac{3}{2}}\left(\frac{k^{\frac{3}{2}}}{Q_2} - \frac{k}{Q_1}\right)\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{7N_2k^3 + 5N_2k^3}{21\eta Q_2} - \frac{2N_2}{3\eta Q_1}k^{\frac{5}{2}}\right.$$
$$\left. + \frac{7N_2Q_2k^2 - 5N_2Q_2k^2}{21\eta Q_1^2}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{4N_1}{7\eta Q_1} - \frac{2N_2Q_2k}{21\eta Q_1^2} + \frac{2N_2k^{\frac{3}{2}}}{21\eta Q_1}\right]\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$= -\frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4k^{\frac{3}{2}}}{7}\left(\frac{k^{\frac{3}{2}}}{Q_2} - \frac{k}{Q_1}\right)\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \left[\frac{4N_2k^3}{7\eta Q_2} - \frac{2N_2}{3\eta Q_1}k^{\frac{5}{2}} + \frac{2N_2Q_2}{21\eta Q_1^2}k^2\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \left[\frac{4N_1}{7\eta Q_1} - \frac{2N_2Q_2k}{21\eta Q_1^2} + \frac{2N_2k^{\frac{3}{2}}}{21\eta Q_1}\right]\frac{df_1(x)}{dx}\Big|_{x_1}$$